

Complexity, Hypersets, and the Ecological Perspective on Perception-Action

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Abstract

The ecological approach to perception-action is unlike the standard approach in several respects. It takes the animal-in-its-environment as the proper scale for the theory and analysis of perception-action, it eschews symbol based accounts of perception-action, it promotes self-organization as the theory-constitutive metaphor for perception-action, and it employs self-referring, non-predicative definitions in explaining perception-action. The present article details the complexity issues confronted by the ecological approach in terms suggested by Rosen and introduces non-well-founded set theory (hyperset theory) as a potentially useful tool for expressing them. The issues and the tool are brought to focus in the concept of affordance that is the basis for explanation of prospective control of action in the ecological approach.

Keywords

affordance, Brusselator, complexity, ecological psychology, James J. Gibson, hyperset, Robert Rosen

1. A Perspective on the Ecological Approach to Perception-Action

The ecological approach to perception and action stands apart from orthodox approaches in two important respects. First, it argues that perception and action are not species of computation (Gibson 1979; Turvey et al. 1981). Second, it insists on the radical proposition that explanation be inferentially strong. Why-questions (why component X? why property Y? why process Z?) should be answerable within the explanation's inferential structure rather than depending on stipulation and hypothesizing (e.g., Shaw and Turvey 1981). This insistence has been frequently phrased in terms of prohibiting "loans of intelligence"—interpreted as ascribing to components, properties, and processes intentional characteristics and specialized knowledge in order to secure explanation.

One consequence of this heterodox position is the proposal by some that the ecological approach ally itself with efforts (e.g., Soodak and Iberall 1987) to extend the traditional strategy of physics to the phenomena of living systems (e.g., Kugler and Turvey 1987; Swenson and Turvey 1991; Shaw and Turvey 1999). The traditional strategy, in a nutshell, is explaining nature through relatively few laws and principles. A vexing issue in this proposed alliance is whether current physics, at its foundations, is fully up to the task (Schrödinger 1944; Rosen 1991; Turvey and Shaw 1995).

The heterodox argument, the radical insistence, and the vexing issue may be all of one piece. Convergence is suggested by Rosen's (1991, 2000) analysis of the question, what distinguishes complex systems from simple ones? The answer he gives depends only secondarily on the system and primarily on the formalism that suffices to model the system. In preview, the answer is that a physical system is *simple* if and only if the behavior of the system is expressible by a model that is computable. Otherwise, the system is *complex*. "Computable" here is Turing-computable. So, for Rosen, a system is complex just in case its models are non-computable. Rosen's answer is crafted in terms of the *modeling relation* depicted in Figure 1.

2. The Modeling Relation

The modeling relation, when realized, depicts what we mean ordinarily by natural law: a pattern of formal entailments that is congruent with the pattern of a system's causal entailments. Formally, for P to entail Q means that Q is a logical or necessary consequence of P . It is a relation such that the truth of P (however defined) ensures the truth of Q . In a formal system, an entailment is a strict, logical or analytical implication. In a natural system, entailment is causality—the relations among events that yield the system's characteristic forms of persistence and change. Examining Figure 1, a formal system is entitled to be called a *model* when the entailments covered by encoding, inference, and decoding yield the same results as

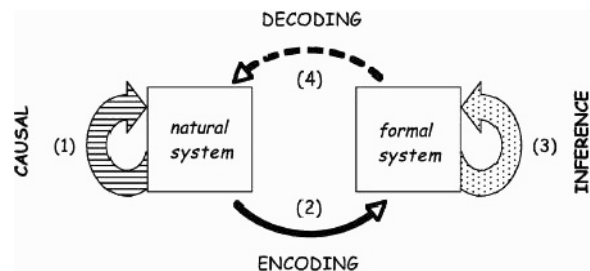


Figure 1.

Modeling relation between causal entailment in a natural system and entailment in a formal system. There are two paths: (1) and (2) + (3) + (4). Path (1) is causal entailment in the natural system. In the second path, (2) encodes phenomena in the natural system to propositions in the formal system, (3) generates theorems entailed strictly by the propositions qua hypotheses, (4) decodes theorems as predictions about the natural system. The formal system is a model of the natural system if the same consequences (answers) are reachable by either path. The modeling relation may be viewed as an embodiment of the notion of *natural law*. (Adapted from Figure 3H.2 in Rosen 1991.)

the causal entailment. To be underscored is the important fact that the model is of causes in the natural system. The model is not of the natural system's behavior.

2.1. Newton's Strategy

The general strategy for constructing a model aimed at satisfying the congruence required by Figure 1 is inherited from Newton. The key elements in the strategy are the concepts of state and recursion. A particle's state is characterized by its position x at time t , that is, $x(t)$, and by its motion at time t as given by the successive time derivatives of $x(t)$: $\dot{x}(t)$, $\ddot{x}(t)$, $\dddot{x}(t)$, and so on, ad infinitum. Recursion pertains to a process that is inherently repetitive, with the outcome of each repetition typically depending on the outcome of the previous repetition. More specifically, a recursively defined function f is one in which successive values are obtained by applying a fixed operation or mapping T to its preceding value. An example is $f(n) = T^n(r)$, where r is an initial arbitrary value and n is the number of times (the n th instance) the operation has been applied.

The essence of Newton's concept of state is its recursive character (Rosen 1991): a particle's state in the future must be entailed from the particle's state now. The instant-to-instant change in a state variable composes a chronicle or time series with as many chronicles as there are state variables. Together, the chronicled variables defining a particle's evolving state form a recursive set $\{x^{(k)}(t)\}$. This is Taylor's theorem that a function can, under certain conditions, be represented as a power series in the neighborhood of a point where it can be differentiated infinitely. The various order derivatives evaluated at that point are the coefficients of the series.

The potentially very large number of quantities defining the state of a single particle (recall the ad infinitum above) prohibits the same state description for the particle's

environment. Clearly, if the environment is all other particles, giving a parallel description of the *state* of the environment is unthinkable. Newton's resolution of this dilemma was to express environment by its effects on the particle, more precisely, by the particle's change of state. Newton embodied the concept of a system's environment in the concept of the force F acting on the particle or system of mass m . From the second law, $F(x, \dot{x}) = m\ddot{x}$, four things follow. First, the particle's state is reduced from infinity to two variables $(x, \dot{x})$ often referred to as the *phase* of the particle. Second, the environment's effects are proportional to the particle's acceleration $\ddot{x}$. Third, the environment's effects are functions of the particle's state defined by position and velocity, that is, the phase $(x, \dot{x})$. And fourth, the environment's effects on the particle are modulated by the parameter m .

To summarize, and to use Rosen's (1988, 1991) terms, environment is encoded as force and system is encoded into a formalism in which the only entailment is a recursion rule such that, given the phase at t_0 ,

$$(x_0, \dot{x}_0) \longrightarrow (x_0 + \dot{x}_0 dt, \dot{x}_0 + F dt). \quad (1)$$

This encoding of natural system has been of great significance to the modeling relation. First, it imposes upon the study of natural systems a strong limitation, namely, the restriction of causal entailment to a sequence of state transitions. Second, it identifies the causal categories with logically independent aspects of the system's description. Third, it accepts as non-entailed the influences that arise from outside the system. Fourth, it limits causation to efficient causation, making ascription of formal or final cause otiose.

2.2. Formalizable Systems

Newton's modeling tactic bears importantly on the conventional understanding of the property, "formalizable." In the most general sense, a formalizable system is one that is expressible in a strictly syntactic manner where each inferential step is a matter of rote symbol manipulation. It is a system that can be modeled by a mathematical or Turing machine.

A Turing machine's parts are symbols that come in different sizes (strings of symbols) and shapes (syntactic roles) and exhibit different motions (transformations by rule). The machine's meaningless symbols are analogous to particles in Newton's framework and its symbol strings are analogous to configurations of such particles. Further, the manipulation of symbols by fixed rules (of arithmetic) that are external to the symbols is analogous to the changes in the states of particles by laws (of forces) that reside in the environment of the particles. At root, the mathematical machine and the Newtonian formalism for material systems are identical (Rosen 1988, 1991, 2000). Recursion is central to both. That is, for both, con-

version of the present to the immediate future exhausts their entailment structures.

The issue of formalizability arose historically in response to actual and threatened paradoxes and inconsistencies in mathematical systems. Such unwelcome features would not occur, it was argued, if the systems were formalizable. Implementing the requirement means ensuring that everything is *predicative*. In an approximate sense, predicativity is used to describe approaches to mathematics in which concepts are introduced in bottom-up fashion. In a slightly more precise sense it is used to describe (a) the construction of all axioms or propositions from definitions that are context-independent and (b) the systematic conversion of all semantic aspects to syntactic forms.

The understanding of predicativity is helped by consideration of its opposite. A non-predicative definition—an impredicativity—is one that refers to a presumed totality of which the object being defined is itself to be a member (e.g., Feferman 1998). It amounts to referring to the whole when defining the parts that will compose the whole or referring to the system when defining the subsystems that will compose the system. In Kleene's (1950: 42) words: "What is defined participates in its own definition." Achieving predicativity then requires rigorously adhering to the guideline: *Disallow any set S that contains members m definable only in terms of S, or members m involving or presupposing S*. This guideline is the *Vicious Circle Principle* suggested by Poincaré (1906). The recommendation, in brief, is: "prohibit impredicative loops."

2.3. Gödel's Lesson

For Rosen, the primary lesson of Gödel's incompleteness theorem is the failure of formalizability. Gödel's theorem states that in any consistent system able to produce simple arithmetic there are formulae that cannot be proved within the system but which are seen to be true from outside the system. There are a number of fairly diverse readings as to what the theorem implies for the scope of human understanding and the theory of mind. Rosen's (1991, 2000) reading is that the theorem pinpoints a fundamental inability to remove all the self-referring impredicative loops of a formal system and it suggests that, rather than being generic, fully predicative mathematical systems are special or non-generic. The generalization of this reading is that simulable or computable systems are rare.

2.4. Causal entailment

Returning to the modeling relation, and the confluence of Newtonian and Turing formalisms: congruence between a natural system's causal entailment and the entailment structure of a fully predicative (fully syntactic) formal system is only possible if the natural system's causal entailment is restricted to "next state" pursuant to "present state." If a natural system's causal entailment is not so restricted then the natural system

and the formal system will be incongruent. It is this incongruity that marks a natural system as not simulable, not computable, and, therefore, complex.

The enrichment of a natural system's entailment would be in the form of causal counterparts of impredicative loops. There are particularly strong indications of such counterparts in the manner in which certain systems assemble themselves and certain systems acquire their properties by way of a larger system that they compose. Systems with causal analogues of impredicative loops exhibit *emergent properties*, *context-dependence*, or both.

Autocatalytic systems are potential examples of systems that self-assemble impredicatively. In chemical autocatalysis, a product of a chemical reaction is a catalyst of the very same reaction that produces it. A key feature of such systems is that they tend to have emergent properties, system-level properties that both arise from and constrain the interaction of the system's components. Being autocatalytic is an emergent property—it is a property of the system as a whole that depends upon and determines the relations among the chemical reactions that make up the system. Consider for example, the autocatalytic Belousov-Zhabotinsky (BZ) reaction. The BZ reaction is a reaction of cerium sulfate, malonic acid, and potassium bromate dissolved in sulfuric acid. When constantly stirred, the reaction forms a chemical clock, a self-assembling oscillator in which the solution changes color in a regular temporal pattern. When the solution is not constantly stirred, the solution forms spiral patterns. These oscillations and spiral patterns are emergent properties of the system, a result of autocatalysis. (See Nicolis and Prigogine 1989.)

Pluripotent stem cells provide an example of a system that acquires its properties in virtue of a larger surrounding system. Stem cells are indeterminate in function until they are inserted into a larger system. When placed in distinct physiological systems, stem cells develop local morphologies and perform local functions appropriate to the distinct systems. In form and function, stem cells are context-dependent. They are beneficiaries of the flow of causal entailment from system to subsystem.

3. Impoverished Entailment and Perception-Action

The heterodox argument, radical insistence, and vexing issue identified in Section 1 can be viewed as reflections upon inadequacies of the modeling relation. Primary conceptions of the ecological view of perception-action seem to be outside the scope of established assumptions about entailment, both causal and formal. Explication and comprehension of direct perception, information in the specificational sense, control without controllers, and affordance, for example, are thwarted by a poverty of entailment (Turvey 2004a). The consequence of this poverty is that the modeling relation for perception-action—

the expression of its laws—is not fully attainable within the contemporary Newtonian scheme.

A reasonable supposition at this juncture is that ecological psychology's subject matter is complex in Rosen's sense, meaning that central ecological notions such as affordance refer to patterns of causal entailments that encode, partially or in full, as impredicative loops. Accordingly, persistent investigation of these key notions will require a method for expressing impredicative definitions and for studying the inferential entailment structures to which they give rise. The theory of hypersets (Barwise and Etchemendy 1987; Aczel 1988; Barwise and Moss 1996) may provide one such method (Kerckel 2002).

4. Primer on Non-Well-Founded Sets (alias Hypersets)

During the late 19th and early 20th century work that codified and established set theory it became evident that certain paradoxes threatened the foundations of set theory. The most famous of these paradoxes was first discussed by Bertrand Russell (1902), and is now known as *Russell's Paradox*. Imagine the barber who shaves all and only those who don't shave themselves. Who shaves such a barber? It turns out that if the barber shaves himself, he does not shave himself, and vice versa. Put in terms of sets, the paradox is as follows. Call all the sets that do not contain themselves as members *normal sets*; those that do contain themselves are *abnormal sets*. Consider *S* the set of all normal sets. Is *S* normal? If so, *S* is a member of the set of all normal sets. But since *S* is the set of all normal sets, *S* would then be abnormal. If *S* is abnormal, then *S* is a member of itself. But if *S* is a member of itself (the set of all normal sets), then *S* must be normal. So: *S* is a member of itself just in case it is not a member of itself, or, equivalently, *S* is normal just in case it is abnormal. This means that the sentence "*S* is a normal set" is both true and false.

The latter was taken to be a serious problem with the theory of sets. Poincaré's (1906) banishment of impredicative definitions was an attempt to solve this problem. "Abnormal set" and "the barber who shaves all and only those who do not shave themselves" are defined impredicatively, so outlawing impredicativities makes defining these types of sets impossible. This solves the paradox, or at least defines it out of existence.

The elimination of abnormal sets leaves us with standard set theory, as taught in college-level logic and mathematics courses. This set theory is *well founded*. There are two related senses in which standard set theory is well founded. First, it is well founded in that its implementation does not lead to paradoxes; that is, it is logically well founded. The second way in which it is well founded comes from graph theory, a standard tool used to understand set membership. In graph

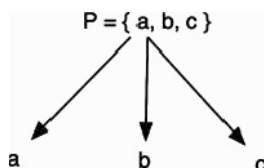


Figure 2.
Diagram of the set $P = \{a, b, c\}$

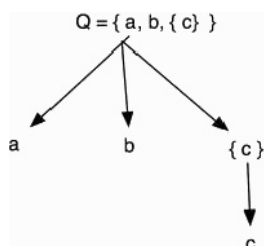


Figure 3.
Diagram of the set $Q = \{a, b, \{c\}\}$

theory, every set determines a tree-like graph in which the set itself is the upper-most node and each member of the set is a descendent node with an arrow pointing to it from the original set of which it is a member; this is applied recursively. Figure 2 shows a graph for the set $P = \{a, b, c\}$. Notice that there is an arrow from the node representing the set P to each of its members. Figure 3 shows a graph for the set $Q = \{a, b, \{c\}\}$. Notice that in this graph, there is an additional complexity in which the node for the member of Q , $\{c\}$, has an additional arrow to its member the object c .

The sets P and Q are well-founded sets; they are not abnormal and do not contain themselves as members. The graphs of P and Q show a property common to the graphs of all well-founded sets. Graphs of well-founded sets contain no infinite cycles or loops, and each path from the original set terminates in some object (its foundation, as it were). In fact, Aczel (1988) proved that a graph will contain no cycles or loops if and only if it is well founded. This means that a graph containing loops or cycles expresses a non-well-founded set. Their presence indicates that some set has itself as a member, or that the concept, system, or definition it models is impredicative.

Consider, as an example, the abnormal set $A = \{A, b, c\}$. This will have the graph in Figure 4. Aczel (1988) also defines the special abnormal set Ω as the set that contains only itself, $\Omega = \{\Omega\}$. Ω is graphed in Figure 5. The sets A and Ω are not allowed in well-founded set theory, but they do have coherent and useful graphs. Aczel's Anti-Foundation Axiom, which embraces non-well-founded (i.e., abnormal) sets is as follows: every graph, loopy or not, pictures a genuine set. Hypersets are defined as graphable sets. So well founded sets and non-well-founded sets are both types of hyperset.

Non-well-founded sets were banished from set theory in the interest of the logical soundness of the theory, and with the

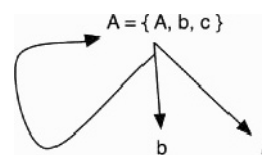


Figure 4.
Hyperset diagram of abnormal set $A = \{A, b, c\}$

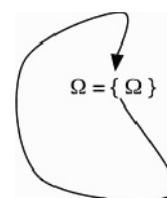


Figure 5.
Hyperset diagram of abnormal set $W = \{W\}$

explicit goal of formalizing mathematics by reducing mathematics to set theory. Re-introducing non-well-founded sets into our set theory makes such formalization impossible. But this loss (if it really is a loss) is offset by the increased ability to model real-world phenomena. Work by Gupta (1982), Barwise and Etchemendy (1987), Gupta and Belnap (1993) and Barwise and Moss (1996) makes clear that many concepts and real-world systems are circular or otherwise not-well-founded, hence illegal according to standard set theory.

The best-known example is the concept “truth,” and the predicate “is true.” Gupta (1982) argues that the semantical paradoxes involving truth, such as the liar (“This very sentence is false”) and the truth-teller (“This very sentence is true”) show that the predicate “is true” can only be defined circularly, and develops a logical technique (revision theory) for dealing with circular concepts. Later, Barwise and Etchemendy (1987) used hypersets to understand the truth predicate. Another example from Quine (1963) is “the most average Yale student.” The identity of this individual is only determined with respect to all the other current Yale students.

For present purposes the more relevant examples are functional properties of a system or subsystem. The stem cell is an example of how context-dependence allows the very same lower-level structure to implement different higher-level objects in different contexts (McClamrock 1995). A mechanical example is found in the carburetor. The very same physically characterized airflow is a choke when it occurs above the fuel jets and a throttle when it occurs below the fuel jets. For a valve, being a choke or a throttle is context-dependent (McClamrock 1995). Both examples, it should be stressed, underscore that impredicative definitions are not limited to the logical or linguistic systems in respect to which they have been most frequently discussed. It should also be emphasized that both examples express a separation of two distinct characterizations of a functional subsystem, one descriptive, one interpretive

(Cummins 1975). The subsystem can be described in terms of locally determined, intrinsic, descriptive properties and it can be described in terms of properties that are non-locally determined, relational, and interpretive—the properties it has by virtue of its role in the embedding system (McClamrock 1995).

In each of the above cases, some object in a system is defined impredicatively, and is therefore illegal in standard set theory. Of course, the illegality of these systems and concepts in standard set theory does not change their behavior, or cause them to cease to exist. Such systems do exist. Adequately modeling them requires a richer set theory, such as hyperset theory, that allows and accounts for the behavior of systems with impredicativities.

The graph theory that goes with hyperset theory will be especially important in what follows. As Aczel has proved and Figures 2–5 demonstrate, graphs of hypersets can be used to determine whether the set pictured is or is not well founded, and if it is not well founded exactly what it is that makes it non-well-founded. These graphs, that is, can be used to determine whether systems contain impredicativities, and so are complex. The graphs also allow us to determine the precise source and nature of the complexity in such systems. We will use this feature of hypersets and their graphs to look at, in order, Rosen's model of living things, the Belousov-Zhabotinsky reaction, and differing conceptions of affordances. Applying hypersets to these three phenomena will shed light upon the systems and show that we can use hypersets to detect and analyze complexity in biological, chemical, and psychological systems.

As a prelude to the discussions of hypersets and their application to various natural systems, an important caveat is in order. Rosen's discussion of the modeling relation makes sense of the relationship between formalisms used by scientists and their relation to what they model. An important contention of this paper, however, is that Rosen's work on the modeling relation is not appropriate as a description of the relationship between animals and reality. On the view defended in this paper, and on the Gibsonian view that underlies it, perception and cognition are not matters of modeling the environment. Ordinary perception and cognition, that is, are not at all analogous to formal theorizing. Rather, they are unmediated contact with the environment. As will be discussed in Section 7 below, perceiving animals and perceived environments are inseparable, and only intelligible in terms of one another. Rosen was something like a transcendental idealist; the ecological approach to perception-action is a form of direct realism. The use of hypersets in elaboration and application of Rosen's views on modeling and complexity in the following sections is intended as demonstration that certain formalisms are inadequate for certain subject matters, and not as endorsing of Rosen's epistemology.

5. Rosen on *Life Itself*

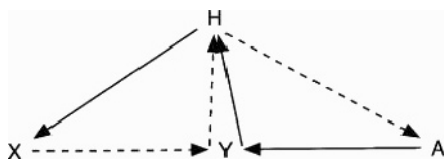
In *The Critique of Judgment* (1790/2000, sections 64–66), Kant argued that an organism is not a machine. His grounds essentially were those of self-organization. Machines neither produce nor reproduce themselves. We elaborate Kant's machine-organism distinction in the following paraphrase of his argument.

For any given thing that we recognize as a machine, for example, a clock or an airplane, the following characteristics seem to hold: The parts of the thing exist for, but not by means of, each other. The parts act together to meet the thing's purpose; their actions, however, have nothing to do with the thing's construction. The thing and its parts rely upon efficient causes arising from outside themselves for their origin and function.

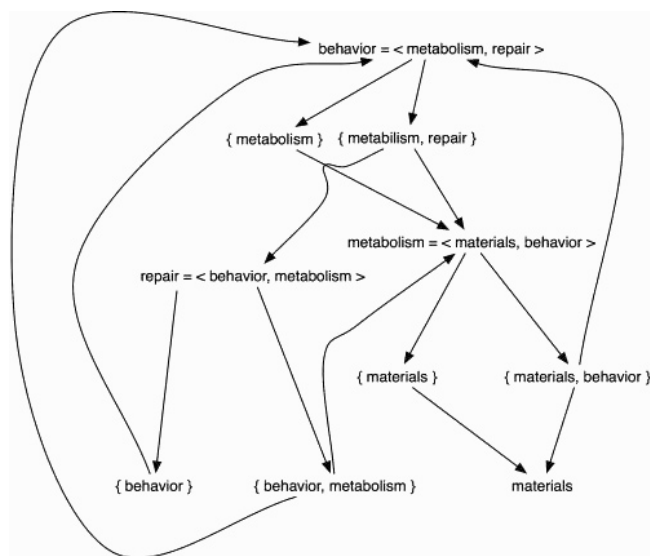
For any given thing that we recognize as an organism, for example, a fly or a tree, the following characteristics seem to hold. The parts of the thing are both causes and effects of the thing; they are not only the means but also the ends. The parts construct and maintain themselves as a unity, each existing by virtue of, and for the sake of, the others and the whole. The thing and its parts are themselves the source of the efficient causes for their origin and function. It is this self-causing nature of living things that led Kant to describe them as having "natural ends" or Aristotelian final causes and "inner purposefulness," and as such requiring teleological explanation.

It is helpful to see Rosen's enterprise in *Life Itself* and *Essays on Life Itself* as an effort to reveal the basis for the machine-organism distinction and to establish the legitimacy and primacy of the strange kind of causality ("unknown to us") that Kant saw as the hallmark of organism. According to Rosen, it is a necessary condition on living systems that their models contain three functions: a metabolic function,¹ a repair function, and a replication function. We will discuss these in order. The metabolic function h_i is a mapping from raw materials $x_i \in X$ to behavior $y_i \in Y$. It determines the way the system reacts to environmental input. A living system must also have a repair function that repairs (sustains) the metabolic function to keep the system reacting appropriately to the environment. That is, it must have a function A which maps behaviors $y_i \in Y$ onto the members of set H , where metabolic $h_i \in H$. Finally, the repair function must itself be maintained. This repair is done by the system's behavior. That is, each member y_i of the set of behaviors Y is itself a replication function that maintains the repair function, by mapping h_i onto $a_i \in A$.

The preceding is, admittedly, complicated. The idea, though, is rather simple. The living system must be metabolizing consistently, so must maintain the causal processes that it uses in metabolizing. This is the function (i.e., natural end or *final cause*) of the repair function. But the repair function must also be maintained, and this is the function (natural end,

**Figure 6.**

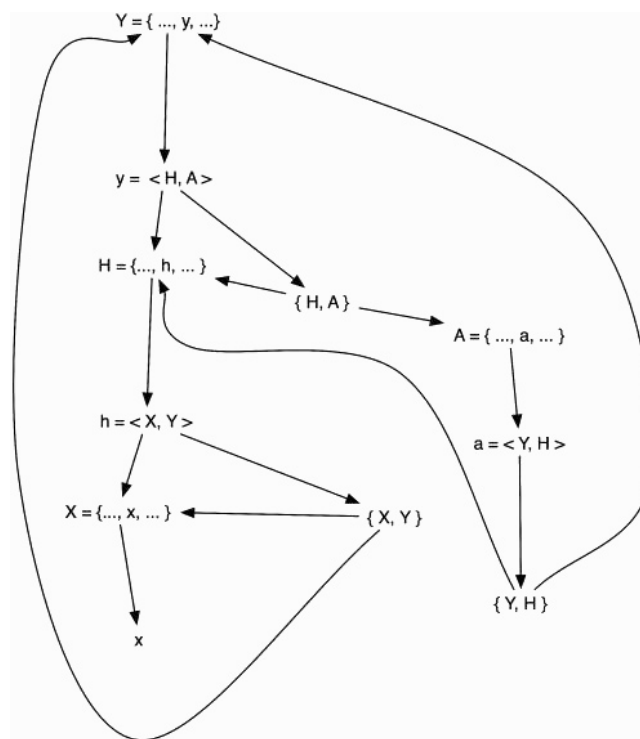
Rosen's model of living systems (1991). Note that $Y = H(X)$ is represented by a solid line from H to X and a dashed line from X to Y.

**Figure 7.**

Simplified hyperset diagram of Rosen's model of living systems. Note that functions are represented as ordered pairs composed of the domain and the range. That is, $a = f(b)$ is represented as $f = \langle a, b \rangle$.

final cause) of the system's behavioral output. Rosen's own diagrammatic representation of his model of living systems, based in Category Theory (e.g., Lawvere and Schanuel 1997), is shown in Figure 6. Note that in this diagram a function P from Q to R (i.e., $P[Q] = R$) represented as " P " connected to " Q " with a solid arrow, and " Q " connected to " R " with a dashed arrow.

Figures 7 and 8 are hyperset diagrams of Rosen's model of living systems. In both figures, two common set-theoretic conventions are followed. First, functions are represented as ordered pairs. That is, if P is a function from Q to R , P is represented as $P = \langle Q, R \rangle$. Second, to preserve their ordering, ordered pairs are represented as sets containing the set containing their first member and the set containing both their members. Thus $\langle Q, R \rangle$ is represented as $\{\{Q\}, \{Q, R\}\}$. The primary difference between the two diagrams is that Figure 7 suppresses some complexity that is present in Figure 8. In Figure 7, the sets of inputs X , behaviors Y , metabolic functions H , and repair functions A are represented as if they were individuals (and not sets of individuals). This is done to make the already-complicated diagram simpler. Figure 8 is presented only to show that doing so makes no relevant difference to our analysis. (It is hoped that providing the fully complex diagram

**Figure 8.**

Hyperset diagram of unsimplified Rosen's model of living systems.

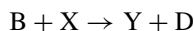
once will allay fears that the simplifications hide anything relevant.)

Of importance in Figures 7 and 8 is that in both cases there are loops, indicating that the sets are not well founded. In each figure there is only one location where the system does not loop back on itself: the set X of inputs from the environment. This makes sense of two important aspects of Rosen's model of life. First, Figure 7 depicts the model of a complex system. It involves irremovable impredicative loops. Second, the fact that the only place the diagrams "bottom out" is with raw materials is also important. Closure to efficient cause is visible in Figure 7. In that diagram, only X , the raw material input from the environment, is not both (1) itself a function in the system and (2) the output of a function of the system. To put this in Rosen's (and Aristotle's) language, all functions but X are both efficient and material causes in the system. X is only a material cause. The hyperset diagrams make clear that this is the case.

We take it that the hyperset analysis of Rosen's model of living systems is useful for two reasons. First, it demonstrates some of Rosen's claims about his model, viz., that it is non-computable, contains impredicativities, is closed to efficient causation. So, we think that the hyperset diagram is enlightening. Second, and more importantly for our purposes, the hyperset diagrams show that such an analysis is useful and does in fact capture important properties concerning the complexity of models.

6. Autocatalysis: The Brusselator

The BZ reaction described in Section 2.4 is typically studied by means of a simpler model of an oscillating, autocatalytic reaction: the Brusselator (Nicolis and Prigogine 1989). The Brusselator is an imaginary reaction of two reactants A and B to produce two products, D and F, via two intermediaries, X and Y. Here are the reactions that make up the Brusselator:



Notice that the intermediaries X and Y are coupled, and that the second and third steps are autocatalytic. X reacts with B to produce Y and Y is a catalyst for the production of more X. Assuming that the concentrations of A and B are controlled externally, the appearance of X leads to the production of Y, which leads to increased concentration of X, which leads to the production of more Y, and so on. This leads to a self-organized oscillation: if X is red and Y is blue, then pursuant to critical increases in A and B, the mixture will oscillate between red and blue states.

Figure 9 is the hyperset representation of the Brusselator. In the graph, each reaction product and intermediary is taken to be the set of the compounds that react to form it. Thus,

$$X = \{\{A\}, \{X, Y\}\};$$

$$Y = \{B, X\};$$

$$D = \{B, X\}; \quad \text{and}$$

$$F = \{X\}.$$

Figure 9 shows that the sets are not well founded, in particular, because (a) X contains itself and (b) Y contains X and X contains Y. Note that the “loopiness” of the diagram reflects the self-organized oscillation in the Brusselator model itself, described in the last paragraph. This loopiness also indicates that the Brusselator is a complex system.

We have now seen hyperset diagrams applied to an impredicatively defined set of functions (Rosen’s model of living systems) and to a causal analogue of impredicativities (the autocatalytic Brusselator). In both cases, the hyperset diagram sheds light on the nature of the complexity in the systems depicted. In the next section, we use hypersets to examine the complexity of a concept foundational to the ecological approach to perception and action, the concept of affordance.

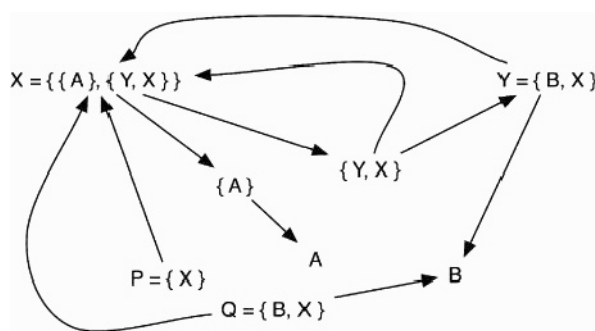


Figure 9.

Hyperset diagram of the Brusselator. Each product is represented as a function of all the sets of reactants that produce it.

7. Affordances

The prime requirement for the behaviors of living things at all scales is that perceiving provide a reliable means for orienting and adjusting activity to the environment. Perceiving is the basis for the practical success of everyday behaviors. It is the basis for flying, swimming, running, jumping, climbing, intercepting, grasping, building, and so forth. For a dragonfly or deer steering its way through the clutter of a forest, the question “to what does the animal’s perceiving refer?” seems to demand the answer “that to which the animal’s activity refers.” The answer equates the proper “objects” of perceiving with the proper “objects” of activity (Gibson 1979; Turvey et al. 1981). To be an “object” of activity is to be an aspect of the surroundings that entails activity. Examples might include a stand on-able surface, a fly through-able opening, a cross over-able gap, and so on. Expressed differently, the examples are a surface that affords standing upon, an opening that affords passage, and a gap that affords crossing. In summary, the perceiving by dragonfly and deer refers to *affordances*—activity-specific meanings of the surroundings. And insofar as flying and running demand prospective (forward looking) control, the perceiving of activity-specific meanings by dragonfly and deer amounts to perceiving the possibilities for action.

Although J. J. Gibson’s introduction of the notion of affordances (Gibson 1977, 1979) left a good deal unsettled about their precise nature, it is very clear that his intention was that affordances were comprehensible only in terms of animal-environment systems. Gibson, for example, spoke of the mutuality of animal and environment, arguing that just as an animal implies a set of affordances it can perceive and use, so too a set of affordances implies an animal that can perceive and use them. The animal and the environment that it perceives, its niche, are defined only in terms of one another. This is evident from Figure 10. Affordances are context-dependent in two different ways. First, the same environmental situation will have different affordances for different animals. Second, the same situation will have different affordances for the same animal, depending upon what the animal is doing. Affordances

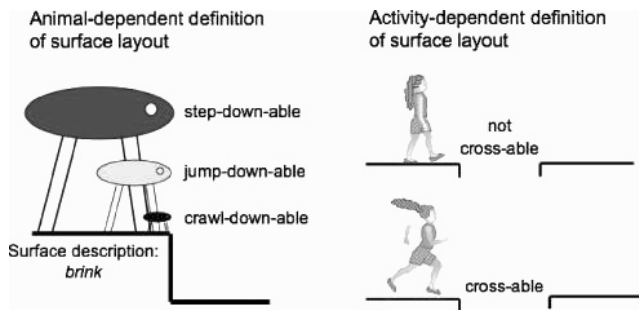


Figure 10.

Affordances are context-dependent. A particular environmental situation will have different affordances for different animals (left), or for the same animal when it is behaving differently (right).

and animals, that is, are intelligible only in terms of their respective contexts, i.e., one another. It would seem, then, that “affordance” would necessarily be defined impredicatively. Yet there has been considerable debate over exactly how one ought to understand affordances.

Historically, there have been two main approaches among ecological psychologists to understanding affordances. Either they are taken to be properties of the environment that have consequences for animal behavior (Michaels and Carello 1981; Turvey 1992; Reed 1996), or they are taken to be relations between animals and environment that have consequences for animal behavior (Sanders 1997; Chemero 2003a,b; Stoffregen 2003). These two views respect Gibson’s intent in developing his ecological psychology. Gibson’s purpose was to develop a theory in which perception is a matter of direct contact with the environment, and to avoid positing intervening mental representations and computations. In contrast, a third view of affordance, proposed by computational cognitive scientists recognizing the need for action-relevant concepts, interprets affordance as a variety of mental representation (Vera and Simon 1993). In this section, we will use hypersets to examine the similarities and differences among these three understandings of affordances. To do so, we focus particularly on three definitions of affordances: Turvey’s (1992) model of affordances as dispositional properties of the environment, Chemero’s (2003b) model of affordances as relations between animals and environments, and Vera and Simon’s (1993) model of affordances as mental representations.

7.1. Affordances as Dispositional Properties of the Environment

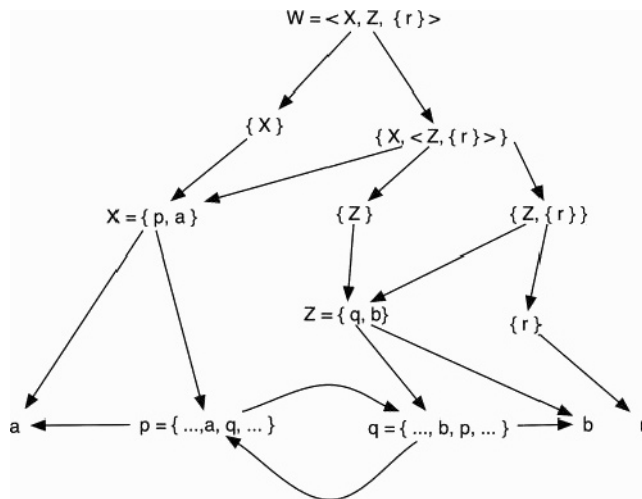
Turvey’s (1992) paper, “Affordances and prospective control: An outline of the ontology,” was intended as a tentative attempt to formalize Gibson’s notion of affordances, and included an explicit invitation for others to refute, expand, or otherwise improve upon his thinking. Turvey (1992: 180) adapted a schema of Bunge’s (1977) to define affordances and their complements, effectivities, as follows:

Let W_{pq} (e.g., a person-climbing-stairs system) = $j(X_p, Z_q)$ be composed of different things Z (e.g., person) and X (e.g., stairs). Let p be a property of X and q be a property of Z . Then p is said to be an affordance of X and q the effectivity of Z (i.e., the complement of p), if and only if there is a third property r such that

- (i) $W_{pq} = j(X_p, Z_q)$ possesses r
- (ii) $W_{pq} = j(X_p, Z_q)$ possesses neither p nor q
- (iii) Neither Z nor X possesses r .

In this definition, j is a joining function that unites the animal and environment into a single system and the property r is the actualization of the afforded behavior. The key idea is that an animal has effectivities (roughly, abilities) and the environmental objects have affordances. When, and only when, there is the right combination of effectivities and affordances, the behavior r is actualized.

It is important to realize, though, that this does not take into account Turvey’s specification of the type of properties that affordances and effectivities are: they are, Turvey claims, *complementary dispositional* properties. It is best to make sense of this claim in the light of Turvey et al. (1981). Two claims that Turvey et al. make about dispositions are especially relevant. First, they claim that dispositions are grounded in, but not identical to, occurrent properties; second, they claim that dispositions are definable only in terms of complementing dispositions. Applying Turvey et al.’s claims about dispositions to Turvey’s definition of affordances, we get the following. Affordances are dispositional properties of environmental objects. Although they have occurrent causal bases, they are definable only in terms of complementing effectivities. Effectivities in turn are dispositional properties of animals that have, but are not identical to, causal bases and are definable only in terms of complementing affordances. We represent this in a hyperset diagram as follows. The system W is represented as the ordered triple $\langle X, Z, \{r\} \rangle$. For simplicity animal Z and environment X are each represented as having just two properties, their dispositional properties and the occurrent properties that serve as the causal bases of the dispositions. That is, $Z = \langle \dots, q, \dots, b, \dots \rangle$ and $X = \langle \dots, p, \dots, a, \dots \rangle$, where b and a are occurrent causal bases, are simplified to $Z = \{q, b\}$ and $X = \{p, a\}$. Affordances and effectivities are inter-defined, and so are represented as sets that include their causal bases and their dispositional complement. Thus, affordance $p = \{a, q\}$ and effectivity $q = \{b, p\}$. (See Figure 11 for the hyperset diagram.) We can see, first of all, that the affordance-effectivity system as understood by Turvey has impredicativities. It is a complex system. The diagram also allows us to understand where the complexity of the system lies. We can see in particular that the only aspects of the system that are “grounded” (not involved in loops) are the occurrent properties: the causal bases of the two dispositional properties and the resultant occurrent action.

**Figure 11.**

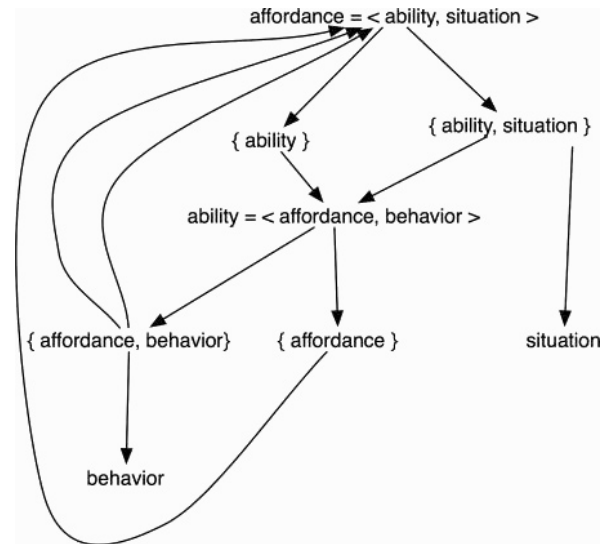
Hyperset diagram of affordances as dispositional properties of environmental objects, as discussed in Turvey (1992).

The complexity, we can see, comes because affordances (e.g., climbable) and effectivities (e.g., ability-to-climb) are defined in terms of one another.

7.2. Affordances as Relations

In contrast to those like Turvey who view affordances as dispositional properties of environmental objects, some ecological psychologists adopt a more plainly mutualist position in which affordances are parts of animal-environment systems (Sanders 1997; Chemero 2003a,b; Stoffregen 2003). Although there are differences among those who hold these positions, we will focus on the view introduced in Chemero (2003a) and developed in detail in Chemero (2003b). According to this view, affordances are relations that hold between abilities of animals and environmental situations. It is important to note that although effectivities are typically considered a technical term for abilities, Chemero (2003b, following Millikan 2000) understands abilities rather differently than Turvey et al. understand effectivities. The primary difference is that Chemero holds that abilities are not dispositions. Instead, abilities are a variety of functional property of animals, in that they are normative and can be exercised well or poorly. Despite this difference from effectivities, abilities are also defined in terms of affordances. An ability is a relation between an affordance and an activity; having the ability to catch a ball means actually being able to catch balls (an activity) when ball-catching affordances are present.

We can understand Chemero's model, then, as saying that token affordances are ordered pairs of particular abilities and particular situations, where those situations are very complex objects that include animals and physical features of the environment. Token abilities, then, are ordered pairs of particular affordances and particular actions. Formally, the affordance

**Figure 12.**

Hyperset diagram of affordances as relations as discussed in Chemero (2003b).

for a behavior B is a set of ordered pairs of particular abilities and situations, Affords-B ($\langle a_1, s_1 \rangle, \langle a_2, s_2 \rangle, \langle a_3, s_3 \rangle, \dots \langle a_n, s_n \rangle$). Similarly, the ability to perform behavior B is a set of ordered pairs of particular affordances and behaviors, Ability-to-B ($\langle \text{affordance}_1, \text{behavior}_1 \rangle, \langle \text{affordance}_2, \text{behavior}_2 \rangle, \dots \langle \text{affordance}_n, \text{behavior}_n \rangle$).

Figure 12 depicts a hyperset diagram of Chemero's model of affordances as relations. Note that, as above, there are simplifications. First, the affordance relation itself is reduced to just one ordered pair, $\text{affordance} = \langle \text{ability}, \text{situation} \rangle$. Similarly, abilities are reduced to a single ordered pair, $\langle \text{affordance}, \text{behavior} \rangle$. The hyperset diagram has a loop connecting affordances and abilities, which are impredicatively defined in terms of one another. This loopiness indicates that the model is not well founded, and so complex in Rosen's sense. Indeed, the only place at which the hyperset diagram "bottoms out" is with the normal environmental conditions for the exercise of the ability and with the activity itself.

7.3. Affordances as Mental Representations

In their paper, "Situated cognition," Vera and Simon argue (contra Gibson and the views just discussed) that affordances are mental representations: "A functional description of the world (i.e., a description in terms of something like affordances) is one that allows simple mappings between our functional models of what is out there (e.g., road curves to the left) and our functional actions (e.g., turn to left)" (Vera and Simon 1993: 21). So an affordance is an internal representation that allows relatively fast mapping between two other representations, one of what the world is like and another of what one might do. Vera and Simon's claim is that what is distinctive about affordances is that they map represented situations onto represented actions without undue deliberation. This, of

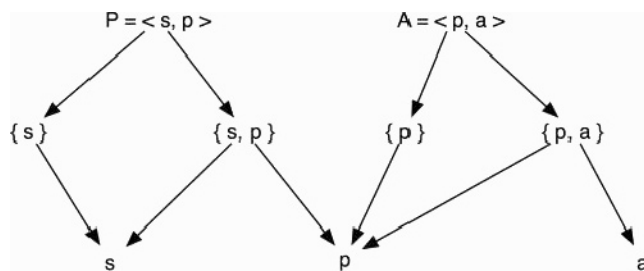


Figure 13.

Hyperset diagram of affordances as representational mappings from perceptual representations to action representations as discussed in Vera and Simon (1993).

course, is in stark contrast to the claims made by ecological psychologists that affordances are environmental objects of direct perception. Vera and Simon take affordances to be representational overlays for perceptual representations, which are themselves the result of a mapping from sensory states. More formally, the idea is that perception is a function from sensory states to perceptual representations, $P = \{ \langle s_1, p_1 \rangle, \langle s_2, p_2 \rangle, \dots, \langle s_n, p_n \rangle \}$. Affordances, then, are mappings from perceptual representations to action representations, $A = \{ \langle p_1, a_1 \rangle, \langle p_2, a_2 \rangle, \dots, \langle p_n, a_n \rangle \}$.

A hyperset diagram of this understanding of affordances is seen in Figure 13. As in earlier diagrams, functions are represented as mappings from just one domain item to one range item. Thus the perception function P is represented as $\langle s, p \rangle$ and the affordance function A is represented as $\langle p, a \rangle$. As is evident in the diagram, these definitions are not impredicative, and the set that satisfies them is well founded. This is unsurprising. Vera and Simon (1993) aimed to formulate a computational account of situated and embodied cognition. Such an account must, of course, be of systems that can be modeled by computers. Impredicativities would introduce a level of complexity into the definition that makes that impossible. Yet in avoiding complexity, Vera and Simon violate Gibson's goal of developing a theory of perception as unmediated contact between an active animal and the environment.

8. Affordance Summary

We have now seen three different definitions of affordance modeled as hypersets. (For ease in what follows, we will call Vera and Simon's view "the computational view," Turvey's "the dispositional view," and Chemero's "the relational view.") We take it that comparing the hyperset models of the computational definition with those of the more properly Gibsonian definitions of affordances makes clear how different the computational view really is. Vera and Simon's definition is designed to bring the perception of affordances under the umbrella of computational cognitive science, and to place them in the head of the observer. The dispositional and relational views do not have these aims, taking affordances to

be genuine features of animal-environment systems. And indeed, these latter definitions of affordances are in many ways similar. We can see at least the following similarities: (1) The hyperset diagrams of both definitions show that they contain impredicativities, hence are complex and unsuitable for computational modeling. This complexity also makes the models difficult to talk about in the computation-inflected language of the cognitive sciences; (2) each definition features circularities between properties of action-enabling aspects of animals and action-enabling aspects of environments; (3) each "bottoms out" only in occurrent or manifest properties (environmental conditions, manifest behavior, bodily features); and (4) both make very clear the complementarity of animal and environment.

This last point of commonality, concerning complementarity, requires some further comment. According to both impredicative views of affordances, there is an important mutuality between animal and the environment it perceives. This mutuality is such that the perceived environment is definable only in terms of the animal and vice versa. In particular, on both the relational and dispositional views, what an animal's environment is depends on what that animal can do. Thus, an animal and its environment are complementary: the animal entails its environment and the environment entails the animals that inhabit it. A consequence of this is that animals with different abilities will, in important ways, inhabit different perceptual environments. We might say that different animals inhabit and perceive different niches, as depicted in Figure 10. We should point out that this is not true of Vera and Simon's computational account of affordances, in which affordances are a mental overlay upon perceptual representations and not genuine, perceivable features of the environment.

The fact that on the dispositional and relational views affordances are defined in terms of abilities, of what animals can do, points to another interesting though rarely noted feature of affordances. Affordances are exquisitely context-dependent and "quicksilvery": they emerge and dissolve very rapidly, due to small changes in the animal, environment or both (Turvey 2004b). The ability to grasp a branch depends, of course, on an animal's hand size, finger strength, and flexibility, and on the circumference of the branch. But it also depends upon the animal's location, the branch's texture, the ground's stability and firmness, and so on—factors that can be changed by a passing shower. In fact, the presence or absence of particular affordances depends upon a spell binding variety of conditions, one or more of which can change in an instant. Hence, because our abilities (or effectivities) are conditional, depending on a wide variety of conditions for their exercise, so too are affordances. Consider crossing a busy street. A walker's maximum stride length, the conditions of the pavement (clean, wet, icy), as well as the layout and velocity of oncoming vehicles (among many other factors) determine

whether street-crossing affordances are present. Yet all of these conditions can change quickly, and, because of this, the experience of missing an opportunity to cross is familiar to all of us. We hesitate, and the street-crossing affordance has disappeared. Yet, usually within moments, another equally fleeting affordance appears. This is a common feature of affordances, and one that is captured by the dispositional and relational views.

This extreme context-dependence, we think, points to another major difference between the dispositional and relational views on one hand and the computational view on the other. The computational view is (*pace* Fodor 2000) fully in line with recent evolutionary psychology. The system that maps affordances onto perceptual representations could, that is, simply be an evolved module. Indeed, one can imagine that the possession of an “affordance module” provides great selective advantage by allowing the animal that possesses it to short-circuit normal action-planning procedures and simply take advantage of vital, time-sensitive environmental opportunities. The dispositional and relational views, on the other hand, point to an environment that changes extraordinarily rapidly from an animal’s point of view. This is true in the guidance of real-time behavior, but also phylogenetically. The presence of animals with particular abilities changes the nature of the niche of those and other animals, which in turn changes the selection pressure on the animals. Thus, one gets a picture of a species and niche over time as coupled dynamical systems, with the nature of the evolving population and the nature of the niche co-evolving over time (Lewontin 1983). This view of the evolutionary process has much more in common with developmental systems approaches (Lewontin 1983; Oyama 1985; Oyama et al. 2003) than with the adaptationist evolutionary biology that is a necessary condition for evolutionary psychology. The context-dependence of affordances, that is, makes ecological psychology incompatible with evolutionary psychology, and a natural ally of the developmental systems approach.

These similarities between the dispositional and relational views, along with the differences between them and the computational view, have implications for the (relatively minor) skirmish among ecological psychologists over whether affordances are dispositional or relational, suggesting the possibility of a rapprochement. Indeed, both Turvey and Chemero understand their views of affordances as claiming that affordances are emergent, relational properties of animal-environment systems (see also Stoffregen 2003). As Turvey et al. (1981), the paper that includes the theory of dispositions Turvey (1992) draws upon, put it: “it is redundant to say . . . that an affordance is a dispositional *and* a relational property as if dispositions were definable independently of their complements” (Turvey et al. 1981: 262). Given the latter, the controversy becomes the comparatively minor question

of which type of impredicatively-defined, relational property affordances are.

9. Conclusion and Future Work

We have seen that hyperset diagrams can sort systems (causal, chemical, definitional) into simple and complex, and, for complex systems, can show the source and nature of the complexity. In the case of Rosen’s model of living systems, we have seen what it is graphically for a system to be closed to efficient causation. When a function in a system is the output of some other function within the system, the hyperset diagram will show that function as failing to “bottom out”. When the system is closed to efficient cause, i.e., when *all* the system’s functions are outputs of other functions within the system, none of the system’s functions will bottom out. In the case of the Brusselator, we have seen something similar for autocatalytic reactions within a system. When we draw hyperset diagrams of reactions, we see loops among the reactants that are involved in the autocatalysis. In the case of affordances, we have seen that the inter-defining of affordances and abilities / effectivities also leads to loopy hyperset diagrams. In effect, these applications of hypersets extend the work of Barwise and Etchemendy (1987) on truth. They showed that hypersets can be used to model natural languages; we have shown that they can be used to model other natural systems.

We close by returning to ecological psychology’s heterodox argument, radical insistence, and vexing issue, discussed in the introduction. Ecological psychology’s heterodox argument, recall, is that perception, action, and cognition are not varieties of computation. The hyperset analysis of affordances shows why this must be the case. When discussed by ecological psychologists, as opposed to computational cognitive scientists who wish to co-opt the notion, “affordance” is defined impredicatively. Affordances are defined in terms of abilities or effectivities, which are in turn defined in terms of affordances. This closed circle of inter-definition means that the animal-environment system, of which both affordances and effectivities are a part, is a complex system. Complex systems, as Rosen has insisted, are not Turing-computable and so not amenable to computational models. Thus, according to ecological psychologists, perception, action, and cognition could not be varieties of computation. The same point accounts for ecological psychology’s radical insistence that causal and definitional entailment in the animal-environment system is system-internal. Because affordances and abilities/effectivities are impredicatively inter-defined, their nature as affordances and abilities/effectivities is entailed by other aspects of the system. The existence of the affordance graspable-with-hand-oriented-vertically depends upon the ability to grasp-with-hand-oriented-vertically, and vice versa. Because of this system-internal entailment, ecological psychologists need not

take “loans of intelligence,” passing the explanatory buck to other systems and other scientists. The hyperset diagrams make this clear.

As for ecological psychology’s vexing issue, it is one shared by all the sciences of complexity. The issue is whether the laws of physics, as currently understood, are sufficient to account for the behavior of complex systems. If, as Rosen has argued, current physics is defined in terms of recursion, then current physical laws will be insufficient to account for systems that are not recursively definable. We have just seen three cases of models of systems that contain impredicativities, and hence are not recursively definable. If these are the appropriate models for the systems in question, then current physics will be unable to account for the behavior of living things, autocatalytic reactions, or perception-action. We need not, of course, have delved into such esoteric matters in order for this to be seen. Consider again Kant’s discussion of living things in *The Critique of Judgment*. The tree, Kant reminds us, actively creates the conditions for its own existence: it produces stems, bark and leaves, and these sustain the tree. Kant (1790/2000: 279) said that this kind of activity has “nothing analogous to any causality we know.” Understanding this kind of activity is the job of the sciences of complexity. We hope to have shown that hypersets will be a useful tool in this endeavor.

Note

1. This inclusion of metabolism as a necessary condition for a living system is echoed by Boden (1999, 2000), and Moreno and Ruiz-Mirazo (1999); see also Shapiro (2006). Boden takes this as implying that Strong ALife is impossible, again in agreement with Rosen.

Acknowledgments

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